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Lehrbuch der Algebra
Working English Translation Batch 03
Further Results on Composition Series; Metacyclic Groups

Translator's note

This working English translation is prepared from the cleaned Kimi-typeset Weber source currently in hand. It continues the group-theory opening of the second volume and covers §§7–8 together with the short connecting appendix paragraph. Mathematical notation has been normalized only lightly.

§7. Further results on composition series

We derive first several important consequences of Jordan's theorem.

Let P be a group and Q a normal subgroup of P , and suppose that P and Q have composition factors

$$A_1, A_2, \dots, A_\lambda \quad \text{and} \quad C_1, C_2, \dots, C_\mu.$$

Then the quotient group P/Q has composition factors

$$A_1, A_2, \dots, A_\lambda, C_\mu^{-1}, \dots, C_2^{-1}, C_1^{-1}.$$

For every composition series of P/Q can be completed to one of P .

Corollary 1.1. If P, Q, R are three groups, each of which is a normal subgroup of the preceding one, and if the index of R in P is a power of a prime number, then the index of R in Q and the index of Q in P are powers of the same prime.

§8. Metacyclic groups

A group P is called *metacyclic* if it possesses a composition series all of whose factors are cyclic.

This definition includes all cyclic groups, and also all groups whose order is a power of a prime number. For every group of prime-power order is, by Sylow's theorem, solvable, hence metacyclic.

Metacyclic groups play an important role in the theory of algebraic equations, since the equations whose groups are metacyclic are solvable by radicals.

Appendix: foundations of group theory

The preceding paragraphs contain the foundations of general group theory as they are used in modern algebra. We have treated the definition of a group, the theory of divisors and cosets, normal subgroups and their properties, many-valued isomorphism, and composition series together with Jordan's theorem.

These concepts are of fundamental importance for the further development of algebra. Upon them is based the theory of abelian groups, which is presented in the following paragraphs of the second section.